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# Wear Equation of AA7050 Aluminium Composites Reinforced with Graphite Particles Produced by Stir Casting

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**ABSTRACT:**

*This research investigates the wear behaviour of aluminium-based composites reinforced with graphite particles produced by stir casting. The objectives include evaluating the effect of graphite reinforcement on wear resistance, analysing wear mechanisms and developing a wear equation to predict the wear rate of the composites. The wear rate of composites rose as the proportion of graphite reinforcement increased due to the decrease in material hardness caused by the integration of graphite, which is a comparatively soft substance when compared to the matrix. The presence of graphite particles also resulted in abrasive wear mechanisms, in which the particles served as abrasives, producing additional material loss and increasing the wear rate. The study revealed that higher load and sliding velocities resulted in higher wear rates because they create higher contact pressures and strains at the contact interface, resulting in more severe wear mechanisms and higher wear rates. The developed wear equation, which incorporates the effects of graphite reinforcement, load, sliding velocity and material hardness, can be used to predict the wear rate of these composites.*

**KEYWORDS:**

*Wear behaviour; Aluminium composites; Graphite reinforcement; Stir casting; Wear equation; Tribological properties*

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## 1. Introduction

Composite materials are widely used in various engineering applications due to their superior mechanical properties compared to conventional materials [1]. The most commonly used composite manufacturing techniques are hand lay-up, resin transfer moulding RTM, vacuum bagging, filament winding, stir casting, infiltration and pultrusion. Stir casting is suitable for mass production, which involves the incorporation of reinforcement materials into a molten metal matrix followed by stirring to ensure uniform distribution [2]. The stirring action helps in dispersing the reinforcement materials evenly, leading to improved mechanical properties of the composite. The effect of processing parameters, such as stirring speed and time, on the microstructure and mechanical properties of aluminium-based composites produced using the stir casting method was investigated [3]. The higher stirring speeds and longer stirring times resulted in improved dispersion of reinforcement materials and enhanced mechanical properties of the composites [4-5]. The influence of different types of reinforcement materials, such as silicon carbide and boron carbide particles, on the properties of aluminium matrix composites produced by stir casting are explored [6]. It is found that the type and size of reinforcement materials significantly affect the final properties of the composites, with silicon carbide particles leading to higher hardness and wear resistance compared to carbon fibres [7].

The wear properties of aluminium-based composites reinforced with silicon carbide particles produced by stir casting were studied [8]. The addition of silicon carbide particles improved the wear resistance of the composites due to the formation of a protective layer on the surface. The wear behaviour of stir-cast aluminium matrix composites reinforced with graphite particles was investigated [9]. Wear rate decreased with an increase in the percentage of graphite reinforcement, attributing this improvement to the self-lubricating properties of graphite were observed [10]. The wear behaviour of polymer matrix composites reinforced with alumina particles were investigated and the wear resistance of the composites increased with an increase in the percentage of alumina reinforcement, attributed to the high hardness and strength of alumina particles. Wear mechanisms of aluminium matrix composites reinforced with graphite particles using scanning electron microscopy SEM were observed and found that abrasive and adhesive wear were the dominant mechanisms [11]. The MML approach to studying the wear mechanisms in graphite MMC was employed [12]. It was found that the MML thickness increased with increasing load and sliding distance, sliding velocity and the size and distribution of the reinforcement particles. indicating a correlation between wear behaviour and contact conditions.

The wear rate of the composites increased with increasing temperature, attributed to the softening of the matrix material at elevated temperatures. The wear mechanism changed from abrasive wear at room temperature to oxidative wear at high temperatures [13]. A rotary drum abrasion test to investigate the wear behaviour of polymer matrix composites reinforced with carbon nanotubes at high temperatures was conducted [13]. The results demonstrated that the addition of carbon nanotubes improved the wear resistance of the composites under high-temperature conditions [14]. Wear behaviour of aluminium matrix composites reinforced with silicon carbide particles using different pin shapes in a pin-on-disc wear test was observed and it was found that the wear rate varied significantly with pin shape, with certain shapes exhibiting higher wear resistance than others [15, 16]. pins with sharper edges experienced higher localized stresses and exhibited more severe wear compared to pins with smoother edges [17]. Several studies have investigated the wear behaviour of aluminium-based composites reinforced with various materials, However, there is limited research focusing specifically on the wear behaviour of graphite composites and the development of a wear equation for predicting their wear rate. The objective of the research was to develop a wear equation to predict the wear rate of the composites based on processing parameters and material properties [18, 19].

## 2. Experimental procedure

The process begins with the preparation of the matrix alloy, which involves placing around 1kg of the aluminium alloy AA7050 in a graphite crucible and heating it to around 850°C with an electric stir casting furnace. Graphite reinforcing particles are heated to 250°C for an hour to eliminate moisture. Because aluminium requires a temperature of around 1100°C to thoroughly wet graphite, a flux potassium hexafluoro titanate $K_2TiF_6$, is used to improve wettability at 850°C. This flux combines with aluminium to generate titanium, which coats the reinforcing particles and strengthens the link between the matrix and reinforcement. The heated graphite covered in aluminium foil is added to the molten alloy. The mixture is then mechanically stirred at 750 rpm for 2 minutes. After that, the flux is added to the melt and the mixture is agitated for another 3 minutes. The melt is then poured into prepared mild steel permanent moulds at 400°C to create specimens with the correct dimensions. The wear test samples were prepared under ASTM G99-05 standards and Ducom computerised pin-on-disc tribometer was used for testing. The experiments were conducted by varying Load, sliding velocity, distance, % reinforcement, specimen shape and temperature as depicted in Table 1. The samples were weighed using an electronic weighing device with a 0.0001g precision and the difference in weight before and after sliding was taken as wear rate.

**Table 1: Wear process parameters**

| Parameters | Levels |
|---|---|
| Load (N) | 10, 20, 30, 40, 50 |
| Sliding velocity (m/s) | 5, 10, 15, 20, 25 |
| Sliding distance (m) | 1000, 2000, 3000, 4000, 5000 |
| % Reinforcement | 2,4,6,8,10 |
| Specimen shape | Hexagon 1, pentagon 2, cylinder 3, rectangle 4, bullets 5 |
| Temperature (°C) | 0, 50, 100, 150, 200 |

## 3. Results and discussion

The increase in wear rate with an increase in percentage of graphite reinforcement, as shown in Fig. 1, can be attributed to the following factors. i.e., the decrease in material hardness with the incorporation of graphite. Graphite is a relatively soft material compared to the matrix material and as the percentage of graphite increases, the overall hardness of the composite decreases. This reduction in hardness makes the material more susceptible to wear and abrasion [10]. The presence of graphite particles can lead to abrasive wear mechanisms. As the composite slides against a counter surface, the graphite particles can act as abrasives, causing more material loss and increasing the wear rate [11]. As the applied load increases, the contact pressure also increases, leading to higher stresses at the contact interface. This increased stress can result in more severe wear mechanisms, leading to higher wear rates. The applied load can influence the deformation behaviour of the composite material. Higher loads can cause more deformation in the material, which can affect its ability to resist wear. In some cases, higher loads can lead to microstructural changes in the material, such as increased porosity or cracking, which can further increase the wear rate. At higher loads, the lubricating effect of graphite may become more pronounced, reducing friction and wear [12]. As the sliding velocity increases, the relative motion between the two surfaces leads to higher frictional forces and temperatures at the contact interface. This increased friction and temperature can accelerate wear processes, leading to higher wear rates [13].

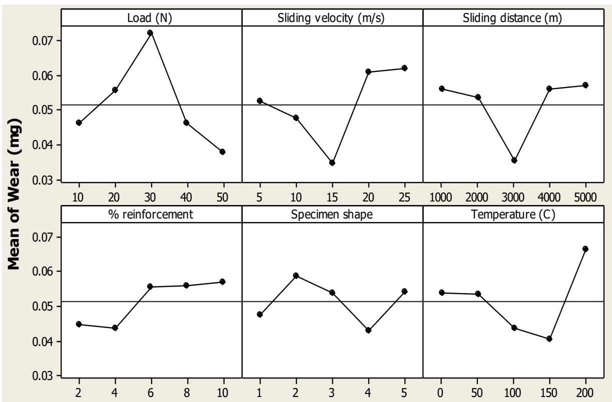


**Fig. 1: Main effect plot of wear rate of AA7050 hybrid composites**

At higher sliding velocities, higher wear debris is generated and interacts with the contacting surfaces, which can lead to more severe wear mechanisms and higher wear rates. Longer sliding lengths cause greater variations in contact circumstances, resulting in variances in friction and wear. This can cause nonlinear wear behaviour, in which the wear rate does not rise linearly with sliding distance. The presence of graphite reinforcement can also affect wear behaviour as the sliding distance varies [14]. Graphite particles can be used as solid lubricants, decreasing friction and wear. However, graphite's efficiency as a lubricant varies with sliding distance, with optimal lubrication occurring at specific distances. At higher temperatures, the material's hardness, strength and ductility vary, potentially affecting its wear resistance. Thermal softening of the material causes greater deformation and material loss, which results in higher wear rates. Temperature changes can affect the behaviour of the lubricants in the composite material. At higher temperatures, the lubricating action of graphite particles become more noticeable, resulting in reduced friction and wear [15].

Different shapes have varied contact areas for a given weight, influencing the distribution of contact stresses and wear processes. Shapes with sharper edges, such as the hexagon and pentagon, have greater localised stresses and more severe wear than ones with smoother edges, such as the cylinder or rectangle. The shape of the specimen can affect the development and behaviour of wear debris. Shapes having complicated geometries, such as pentagons and bullets, trap more wear debris, resulting in greater abrasive wear and wear rates [16]. In contrast, simpler designs with smoother surfaces, such as the cylinder and rectangle, accumulate less wear debris and wear at lower rates. The alignment of edges and corners with the sliding direction affects the degree of material loss.

The Archard wear equation is a fundamental equation used to predict the rate of material removal due to sliding wear. To incorporate sliding velocity, sliding distance, applied load and reinforcement as process parameters into the Archard wear equation [17],

$$\mathrm{V} = \mathrm{k}'.L.P.\mathrm{Vs}.\mathrm{R} \quad (1)$$

Where, V is the wear volume of material removed, k' is the wear coefficient material property, L is the sliding distance, P is the normal load, Vs is the sliding velocity and R is the reinforcement factor. Which accounts for the presence and characteristics of any reinforcement materials in the composite material. This factor can vary based on the type, size and distribution of reinforcements and simulation of loiter and glide. The modified wear coefficient k' can be related to the original wear coefficient k using hardness reduction factor Hr as,

$$k' = k.1 - Hr \quad (2)$$

To incorporate the increase in wear rate with the addition of graphite particles into the wear equation,

$$V = k''.L.P.Vs.R \quad (3)$$

The modified wear coefficient k" can be related to the previously modified wear coefficient k' and the wear rate increase factor Wr due to the addition of graphite particles as follows,

$$k'' = k'.1 + Wr \quad (4)$$

The modified equation would then be [18],

$$V = k''.L.P.Vs/R \quad (5)$$

The adjustment reflects the inverse relationship between wear resistance and the reinforcement factor.

Given that hardness reduces with the incorporation of graphite and wear rate increases with graphite, both factors should be in the denominator of the wear equation according to the modified Archard wear model as,

$$V = k''.L.P.Vs/H.R \quad (6)$$

This equation reflects the combined effects of hardness reduction and increased wear rate due to the addition of graphite particles, both of which contribute to a decrease in wear resistance. To differentiate the wear volume V with respect to the reinforcement factor R, we first need to express V in terms of R in the wear equation [19]. Given the wear equation:

$$V = K''.L.P.Vs/H.R \quad (7)$$

We can rewrite it as:

$$V = 〖k^{\wedge}''.L.P.V〗\_s/H.R \quad (8)$$

Differentiating V with respect to R gives,

$$dV/dR = 〖k^{\wedge}''.L.P.V〗\_s/〖H.R〗^{\wedge}2 \quad (9)$$

Yang equation is given by,

$$〖V = Ae〗^{\wedge} - BL \quad (10)$$

By incorporating sliding velocity Vs, load P and reinforcement R into the wear equation V = 〖Ae〗^-BL, this was to introduce additional constants that account for the effects of sliding velocity, load and reinforcement [20]. The modified equation can be expressed as:

$$V = A^{\wedge}('e^{\wedge}(-B^{\wedge}' L)\, V\_s^{\wedge}C\, P^{\wedge}D\, R^{\wedge}E) \quad (11)$$

A', B', C, D and E are constants that account for the effects of sliding velocity, load and reinforcement, to modify the equation to reflect the negative impact of reinforcement and hardness on the wear volume, by incorporating these factors in the denominator of the exponential term. Here's the modified equation:

$$V = A'e^{\wedge}(-B^{\wedge}' L)\, V\_s^{\wedge}C\, P^{\wedge}D\, 〖/R〗^{\wedge}E\, 〖.H〗^{\wedge}F \quad (12)$$

To modify the equation to $e^{\wedge}(-BLVP)$, it was directly incorporated the sliding velocity Vs and load P into the exponential term. Here's the modified equation:

$$V = A'e^{\wedge}(-B^{\wedge}' LV\_s\, P)\, 〖/R〗^{\wedge}E\, 〖.H〗^{\wedge}F \quad (13)$$

To differentiate the wear volume V with respect to the reinforcement factor R in the equation V=A'e^-B^' LV_s P 〖/ R〗^E 〖.H〗^F, we first need to express V explicitly in terms of R

$$dV/dR = -E.(A'e^{\wedge}(-B^{\wedge}' LV\_s\, P))/(R^{\wedge}(E+1)\, 〖.H〗^{\wedge}F) \quad (14)$$

Equating the two derivatives dR/dV,

$$\mathrm{H}.R^{\wedge}2 - 2.k''.\mathrm{L.P.Vs} = -\mathrm{E}\, 〖.H〗^{\wedge}F.A'.e^{\wedge}(-B^{\wedge}' LV\_s\, P) \quad (15)$$

To find the reinforcement coefficient R in the equation

$$\mathrm{H}.R^{\wedge}2 - 2.k''.\mathrm{L.P.Vs.R} - \mathrm{E}\, 〖.H〗^{\wedge}F.A'.e^{\wedge}(-B^{\wedge}'.L\, 〖.V〗\_(s.)\, P) = 0 \quad (16)$$

This is a quadratic equation in R to be solved using,

$$R = (-b \pm \sqrt{}(b^{\wedge}2 - 4ac))/2a \quad (17)$$

Where, a = H. $b = 2.k''$. L. P. Vs and c = - E $〖.H〗^{F.A'.e^{(-B'.L〖.V〗_{(s.)}P)}}$. To solve the quadratic equation,

$$H.\ R^2 - 2.k''.\ L.\ P.\ V_s\ .R\ -\ E〖.H〗^{F.A'.e^{(-B'.L〖.V〗_{(s.)}P)}} = 0$$

$$R = \left(-k''.L.P.V_s \pm \sqrt{\left(〖k''.L.P.V_s〗^2 + H.E〖.H〗^{F.A'.e^{(-B'.L〖.V〗_{(s.)}P)}}\right)}\right)/H \quad 18$$

This equation represents the solution for R that satisfies the original equation for the wear volume. Depending on the specific values of k", L, P, Vs, H, E, F, A', B' and R, it was used to calculate the reinforcement factor R that corresponds to the given wear volume equation.

## 4. Conclusion

The increase in wear rate as the amount of graphite reinforcement increases can be attributed to a decrease in material hardness and abrasive wear processes generated by graphite particles. Higher loads and sliding velocities offer higher wear rates due to increased contact pressures and strains at the interface, resulting in more severe wear. At increasing temperatures, thermal softening generates more deformation and material loss, resulting in increased wear rates. The developed wear equation integrates the influences of sliding velocity, sliding distance, applied load and reinforcement, resulting in a comprehensive model for predicting composite wear rate. The wear equation shows an inverse connection between wear resistance and reinforcing factor, which reflects the combined effects of reduced hardness and higher wear rate caused by the inclusion of graphite particles. The obtained quadratic equation allows for the computation of the reinforcement factor that corresponds to a particular wear volume equation, making it feasible to optimise the composite material for engineering applications.

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